Module 4

Sampling distribution

Science of statistics is classified as

a) Descriptive

b) Inductive

Descriptive statistics consists of describing some characteristics of the numerical data.

Inductive statistics is the logic of drawing statistically valid conclusions about the totality of cases or items termed as population.

Statistical investigation is based on examining a part known as sample, which is drawn from the population in scientific manner.

Universe / Population : group or individuals under study

Finite population : Population containing finite number of objects or items.

eg:- Students in a college.

Infinite population : Population containing infinite number of objects or items.

eg:- Population of stars in the sky

Existent : Population consists of concrete objects.
eg:- Population of books
Hypothetical : Population which does not consist of concrete objects.
eg:- Population of throws of a die or a coin.

Sampling

Sample : Finite subset of the population, selected from it with the objective of investigating its properties.

Sample size : Number of units in the sample.

Sampling : Tool which enables us to draw conclusions about the characteristics of the population after studying only those objects or items that are included in the sample.

Objectives of the sample

- To obtain the optimum ~~state~~ results.
- To obtain the best possible estimates of the population parameters.

Sampling errors

Error involved in approximations about the population characteristics on the basis of the sample.

Parameter & statistic

Parameters : Functions of the population values.
Statistical constants like mean (μ), variance (σ^2) skewness (β_1), kurtosis (β_2)

Statistic : Functions of the sample observations.
Statistical constants like mean ($\bar{x}$), variance (s^2), skewness (b_1), kurtosis (b_2)

Census versus sample

Census method / complete enumeration

Here 100% of the population is inspected and each and every unit of the population is enumerated.

Sample method / Partial enumeration

Some units are selected from the population and they are inspected.

Errors in statistics

1) Sampling and Non sampling errors

Sampling errors

- Faulty selection of the sample.
- Faulty demarcation of the sampling units.

Non sampling errors

- Non responsive and inadequate or incomplete response.
- Improper coverage
- Compiling errors.

2) Biased & Unbiased errors

Biased errors :

- Bias on the part of the enumerator or investigator.
- Bias in the measuring instrument or equipment used for recording observations.
- Respondents bias.

Unbiased errors :

- Making under-estimate or over-estimate.

TYPES OF SAMPLING

i) Purposive / Subjective / Judgement Sampling
ii) Probability Sampling
iii) Mixed Sampling

i) Purposive Sampling

A desired number of sample units is selected delibrately or purposely depending upon the object of the enquiry so that only the important item representing the true characteristics of the population are included in the sample.

- A serious drawback of this scheme is that it is highly subjective in nature.
- Since the sample selection depends entirely on the personal convenience, belief of the investigator.

ii. Probability Sampling

Provide a scientific technique of drawing samples from the population according to some laws of chance in which each unit in the universe are some pre assigned probability of being selected in the sample.

Different types of sampling are in which

- Sampling units has an equal chance of being selected.
- Sampling units have varying probability of being selected.
- Probability of selection of a unit is proportional to the sample size.

iii) Mixed Sampling

Sampling units are selected partially according to some probability laws.

Probability mixed sampling schemes are

1) Simple random sampling
2) Stratified random sampling
3) Systematic random sampling
4) Multistage random sampling
5) Quota random sampling
6) Area sampling
7) Simple cluster
8) Multistage cluster
9) Quasi random sampling

Simple random sampling

Technique in which sample is so drawn that each in every unit in the population has an equal and independent chance of being included in the sample. If the unit selected in each draw is not replaced in the population before making the next draw is simple random sampling without replacement (srswor)

If the unit selected in each draw is replaced in the population before making the nextdraw is simple random sampling with replacement (srswr).

A random sample may be selected by

i) Lottery Method

ii) Use of random number tables.

i) Lottery Method

It consists of identifying each and every unit in the population with a distinct number which is recorded on a slip. This slip should be homogenous These slips are put in a bag throughly shuffled and the slips drawn one by one. The sampling unit corresponding to numbers in the selected slip will constitute the random sample.

ii) Use of random number tables

The lottery method is time consuming if the population size is very large. In such cases, we are using the help of random number of tables. In This table, each of the digits 0, 1, 2 ... 9

appears with the same frequency and independent of each other. If you want to select a sample from a population of size $N (\leq 99)$, then the numbers can be combined 2×2 to give pairs from 00 - 99.

Steps.

1. Identifying N units in the population.
2. Select at random any page of the random number table and pick up the numbers in any row, column or diagonal at random.
3. The population unit corresponding to the numbers selected constitute the random sample.

<u>Stratified Random Sampling</u>

Stratification means division into layers or groups.

Steps:

1. Stratify the given population into a number of subgroups or subpopulations known as strata such that
   a) the units within each stratum are homogenous as possible.
   b) the difference between n various strata are marked as possible.
   c) various strata are non-overlapping.
2. Draw simple random sample without replacement from each of the 'n' strata.

The sample of $N = \Sigma n_i$ units is stratified random sample (without replacement). Technique of drawing such a sample is stratified random sample.

<u>Allocation of sample size in stratified sampling.</u>

1) Proportional allocation
Items are selected from each stratum in the same proportion as they exist in population.

2) Optimum allocation
Samples are drawn from the various strata is determined by the principle of optimisation.

3) Disproportionate allocation
An equal number of items is taken from every stratum regardless of how the stratum is represented in the population.
A stratified sample in which the no. of items selected from each stratum is independent of its size is called disproportionate stratified sample.

<u>Systematic Sampling</u>
Here only the first sample unit is selected at random and the remaining units are automatically selected in a definite sequence at equal spacing from one another.

<u>Cluster Sampling</u>
The Total population is divided depending on problem under study into some recognisable sub divisions called clusters and a simple

random sample of these clusters is drawn.

Properties

- Cluster should be as small as possible.
- The no. of sampling units in each cluster should be approximately same.

Multistage Sampling

Carried out in various stages. Here the population is regarded as made of primary units each of which is further composed of a number of secondary stage units and so on, till we ultimately reach the desired sampling unit in which we are interested.

Quota Sampling

Special form of stratified sampling.
The quota of the units to be examined by the investigation from the stratum assigned to him is fixed for each investigator.